RF Components: Characterization and Measurement (part -1)

Soumi Mandal, Roll number: 261310613[1, *]

[1] SPASE Department, Indian Institute of Technology, Kanpur, Uttar Pradesh, India

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I. OBJECTIVE

(a) To characterize an RF filter, identify its type, and determine its cutoff frequencies or passband.

(b) To demonstrate a quarter-wave open-circuit stub as a notch filter.

(c) To observe frequency conversion in an RF mixer and estimate its conversion loss.

(d) To measure the power division and insertion loss of a power splitter/combiner.

(f) To estimate the input reflection coefficient and return loss of the filter.

II. INTRODUCTION

Radio-frequency (RF) systems are used to receive, process, and analyse electromagnetic signals over a specified frequency range. In a radio astronomy receiver, the received signals are generally very weak and must therefore be processed carefully so that the desired information is preserved while unwanted signals, losses, reflections, and nonlinear distortions are minimized.

A typical RF receiver consists of several functional blocks. The antenna converts the incident electromagnetic radiation into an electrical signal. A low-noise amplifier (LNA) increases the signal level while attempting to introduce as little additional noise as possible. A mixer translates the signal from one frequency to another using a local oscillator (LO). Filters select the required frequency range and suppress unwanted components. Power splitters and combiners distribute or combine RF signals, while additional amplifiers provide the required signal gain before further processing or digitization by passing it through ADC (Analog to Digital Convertor).

The performance of an RF receiver therefore depends not only on the gain of its individual components but also on their frequency response, impedance matching, insertion loss, and nonlinear behaviour. The characterization of these properties is essential for understanding the behaviour of a complete RF signal chain.

* soumimandal26@iitk.ac.in

III. THEORY

A. Overview of RF Components in a Receiver Chain

A radio-frequency (RF) receiver is designed to detect and process very weak electromagnetic signals over a selected range of frequencies. A typical receiver therefore consists of several RF components, each performing a specific function. The individual components cannot generally be considered independently because their transmission loss, gain, impedance matching, nonlinear behaviour, and frequency response collectively determine the overall performance of the receiver.

A simplified RF receiver chain can be represented as

$$Antenna \rightarrow LNA \rightarrow Mixer \rightarrow Filter \rightarrow LNA \rightarrow Filter \rightarrow ADC. \quad (1)$$

The antenna receives the incident electromagnetic radiation and converts it into an electrical RF signal. Since astronomical signals are generally very weak, a low-noise amplifier (LNA) is often placed near the beginning of the receiver chain. The purpose of the LNA is to increase the signal power while introducing as little additional noise as possible.

The mixer performs frequency translation. It combines the incoming RF signal with a local oscillator (LO) signal and generates new frequency components corresponding to the sum and difference of the two input frequencies. A suitable filter can then select the required intermediate-frequency (IF) component.

B. RF Filter

An RF filter is a frequency-selective component that allows signals within a desired frequency range to pass while attenuating signals outside that range. The transmission characteristic of a filter is determined by measuring the output power as the input frequency is varied.

The four basic filter types considered in RF systems are low-pass, high-pass, band-pass, and band-reject or notch filters.

1. Low-Pass Filter

A low-pass filter transmits frequencies below a specified cutoff frequency and attenuates frequencies above it. The cutoff frequency is denoted by f_c. As the frequency increases beyond the passband, the attenuation increases and the signal eventually enters the stopband.

The cutoff frequency is commonly determined from the -3 dB point of the measured frequency response. The stopband edge can be represented by f_s and indicates the frequency at which the response is considered to have entered the stopband.

2. High-Pass Filter

A high-pass filter has the opposite frequency-selective behaviour. It suppresses frequencies below its cutoff frequency and transmits frequencies above the cutoff. The cutoff frequency f_c is determined from the measured response using the appropriate -3 dB criterion.

3. Band-Pass Filter

A band-pass filter allows frequencies within a specified interval to pass while attenuating frequencies below and above this interval. The lower and upper cutoff frequencies are represented by f_1 and f_2, respectively.

Thus, the principal transmission region can be expressed approximately as

$$f_1 < f < f_2. \tag{2}$$

The frequency at the centre of the transmission band is denoted by f_0.

4. Band-Reject or Notch Filter

A band-reject filter suppresses a selected frequency range while transmitting frequencies outside that range. When the rejected frequency range is relatively narrow, the filter is commonly called a notch filter.

The frequency at which the maximum attenuation occurs is known as the notch frequency, f_0. In an experimental frequency sweep, the notch appears as a pronounced dip in the transmitted output power.

C. Filter Insertion Loss

Insertion loss is an important parameter used to quantify the attenuation introduced by an RF filter. It represents the reduction in signal power resulting from inserting the filter into the RF signal path.

A reference measurement is first obtained without the filter. Let the measured reference input power be P_{in} and the power measured after inserting the filter be P_{out}. When these powers are expressed in dBm, the insertion loss is

$$IL = P_{\text{in}} - P_{\text{out}}. \tag{3}$$

The insertion loss is expressed in decibels.

A small insertion loss indicates that the filter transmits the signal efficiently, whereas a large insertion loss indicates stronger attenuation.

By measuring the insertion loss over a range of frequencies, the frequency response of the filter can be obtained:

$$IL = IL(f). \tag{4}$$

A plot of insertion loss against frequency can then be used to identify the type of filter and its characteristic frequencies.

For low-pass and high-pass filters, the cutoff and stopband frequencies can be identified from the measured response. For band-pass and notch filters, the lower frequency, centre or notch frequency, and upper frequency can be determined from the corresponding features of the measured response.

The -3 dB cutoff frequencies are measured relative to the minimum insertion loss within the passband. This provides a consistent definition of the useful transmission bandwidth of the filter.

D. Input Reflection Coefficient

In an RF system, maximum power transfer requires proper impedance matching between connected components. If the input impedance of a filter is different from the characteristic impedance of the transmission line, part of the incident RF signal is reflected toward the source.

Let Z_{in} represent the input impedance of the filter and Z_0 represent the characteristic impedance of the RF transmission system. The voltage reflection coefficient is defined as

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}}, \tag{5}$$

where V_{inc} is the amplitude of the incident voltage wave and V_{ref} is the amplitude of the reflected voltage wave.

For an impedance discontinuity, the reflection coefficient can be written as

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}. \tag{6}$$

The magnitude of the reflection coefficient, $|\Gamma|$, indicates the degree of mismatch between the filter and the transmission system.

For a perfectly matched system,

$$Z_{\text{in}} = Z_0, \tag{7}$$

which gives

$$|\Gamma| = 0. \tag{8}$$

Thus, no incident voltage is reflected.

For complete reflection,

$$|\Gamma| = 1. \tag{9}$$

Therefore, a smaller value of $|\Gamma|$ represents better impedance matching.

E. Return Loss

Return loss is another parameter used to quantify impedance matching. It is related to the magnitude of the reflection coefficient by

$$RL = -20 \log_{10} |\Gamma|. \tag{10}$$

Return loss is expressed in decibels.

A small reflection coefficient corresponds to a large return-loss value and indicates a good impedance match. Conversely, a large reflection coefficient produces a smaller return loss and indicates greater mismatch.

For an ideal match,

$$|\Gamma| = 0, \tag{11}$$

and consequently,

$$RL \to \infty. \tag{12}$$

For complete reflection,

$$|\Gamma| = 1, \tag{13}$$

giving

$$RL = 0 \text{ dB}. \tag{14}$$

In the experimental arrangement, the reflection coefficient can be estimated from the maximum and minimum voltage amplitudes observed at the measurement junction. Let these voltages be $V_{\max}$ and $V_{\min}$. The incident voltage amplitude can be estimated as

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2}, \tag{15}$$

while the reflected voltage amplitude is

$$V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2}. \tag{16}$$

Therefore, the magnitude of the reflection coefficient becomes

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{17}$$

The return loss can consequently be calculated from

$$RL = -20 \log_{10} \left(\frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \right). \tag{18}$$

This measurement provides an estimate of the input matching because practical effects such as cable loss, connector mismatch, and loading by the measuring instrument can influence the observed voltages.

F. Quarter-Wave Open-Circuit Stub

A transmission-line stub is a section of transmission line connected to an RF circuit to produce a frequency-dependent impedance. In this experiment, an open-ended coaxial stub is connected to the main RF transmission path using a Tee connector.

The electrical behaviour of a transmission line depends on its physical length relative to the wavelength of the signal. When an open-ended stub has an electrical length approximately equal to one quarter of the guided wavelength, the impedance at its input is transformed from a high impedance at the open end to a low impedance at the Tee junction.

The quarter-wave condition is

$$L = \frac{\lambda}{4}, \tag{19}$$

where L is the physical length of the stub and λ is the guided wavelength.

The propagation velocity of a signal inside a coaxial cable is generally lower than the speed of light in vacuum. This is described by the velocity factor V_F. Consequently, the expected notch frequency is

$$f_0 = \frac{V_F c}{4L}, \tag{20}$$

where

- f_0 *is the expected notch frequency,*
- V_F *is the velocity factor of the coaxial cable,*
- c *is the speed of light in vacuum, and*
- L *is the physical length of the stub.*

At the quarter-wave frequency, the low impedance presented by the stub at the Tee junction strongly affects the propagation of the signal along the main path. A large reduction in transmitted power therefore occurs, producing a notch in the frequency response.

The attenuation introduced by the stub can be calculated using

$$A = P_{\text{in}} - P_{\text{out}}, \tag{21}$$

where P_{in} *is the reference power without the stub and* P_{out} *is the measured power with the stub connected.*

The theoretical notch frequency may differ slightly from the experimentally measured value. Practical deviations can arise from the physical construction of the connector, the behaviour of the open end, cable properties, and other non-ideal effects.

G. RF Mixer

An RF mixer is a nonlinear three-port device used to translate a signal from one frequency range to another. Its three ports are generally designated as the RF, local oscillator (LO), and intermediate-frequency (IF) ports.

If an RF signal with frequency f_{RF} *and an LO signal with frequency* f_{LO} *are applied to the corresponding inputs, nonlinear mixing produces new frequency components.*

The two principal mixing products occur at the sum and difference frequencies:

$$f_{\text{sum}} = f_{\text{RF}} + f_{\text{LO}}, \tag{22}$$

and

$$f_{\text{diff}} = |f_{\text{RF}} - f_{\text{LO}}|. \tag{23}$$

When the RF frequency is greater than the LO frequency, the difference-frequency component can be written as

$$f_{\text{IF}} = f_{\text{RF}} - f_{\text{LO}}. \tag{24}$$

The required IF component can subsequently be selected using an appropriate filter.

Frequency conversion is an important operation in RF receivers because it allows a signal received at a relatively high frequency to be converted to another frequency where amplification and filtering may be more convenient.

A practical mixer does not convert all of the applied RF power into the desired IF component. Some power is distributed among other mixing products, while additional losses occur within the device.

If P_{RF} *is the RF input power and* P_{IF} *is the measured power of the selected IF component, the conversion loss is*

$$L_{\text{conv}} = P_{\text{RF}} - P_{\text{IF}}. \tag{25}$$

A lower conversion loss indicates more effective conversion of the RF input into the selected IF output.

H. Power Splitter

A power splitter divides an input RF signal into two or more output paths. A two-way equal splitter ideally divides the available input power equally between its two outputs.

For an ideal two-way splitter,

$$P_{\text{out}} = \frac{P_{\text{in}}}{2}. \tag{26}$$

Expressed in decibels, the power reduction at each output is

$$10 \log_{10}\left(\frac{1}{2}\right) = -3.01 \text{ dB}. \tag{27}$$

Thus, ideally, each output is approximately 3 *dB below the input power:*

$$P_{\text{out}} = P_{\text{in}} - 3.01 \text{ dB}. \tag{28}$$

A practical splitter introduces additional insertion loss because of conductor losses, dielectric losses, impedance imperfections, and other non-ideal effects.

If the powers measured at the two output ports are P_{out1} *and* P_{out2}*, the losses associated with the two paths are*

$$L_1 = P_{\text{in}} - P_{\text{out1}}, \tag{29}$$

and

$$L_2 = P_{\text{in}} - P_{\text{out2}}. \tag{30}$$

Comparison of these two values provides information about the equality of the two output paths and the additional losses introduced by the practical splitter.

I. Power Combiner

A power combiner performs the reverse operation of a power splitter. Instead of dividing one signal into multiple paths, it accepts signals at two or more input ports and combines them at a common output.

The output of a combiner depends on the frequencies, amplitudes, and relative phases of the input signals.

When two signals have different frequencies, the output spectrum contains the corresponding frequency components. When the signals have the same frequency, their relative phase becomes important.

For two signals of the same frequency, signals that are in phase tend to reinforce one another, while signals with an opposing phase relationship can partially or completely cancel.

Consequently, a practical RF combiner is not simply a device that adds two power levels numerically. The phase and amplitude relationships between the input signals must also be considered.

J. Overall Significance of RF Characterization

The characterization of RF components requires consideration of both their desired and undesired effects on a signal.

For a filter, the transmission response determines which frequencies are passed or attenuated, while the reflection coefficient and return loss describe the quality of the input impedance match.

For a quarter-wave stub, the frequency-dependent impedance transformation of the transmission line produces a sharp attenuation at the quarter-wave resonance.

For a mixer, the important behaviour is frequency translation from the RF and LO frequencies to sum and difference frequencies.

For a splitter and combiner, the distribution and combination of RF power depend on the characteristics of the network and, in the case of coherent signals, on their relative phases.

Together, these parameters provide a comprehensive description of the behaviour of the RF components and help determine their suitability for use in an RF receiver chain.

IV. COMPONENTS REQUIRED

- *RF signal generator(s)*
- *Spectrum analyser*
- *Digital storage oscilloscope*
- *Filter*
- *Mixer*
- *Power splitter/combiner*
- *RF amplifier and the required DC power supply*
- *RF Tee connector*
- *Open-ended coaxial stub*
- *Short reference coaxial cable*
- *50 Ω terminations*
- *Suitable RF cables, adapters, and attenuators*

V. EXPERIMENTAL SETUP

A. General Setup

The basic transmission measurement consists of an RF signal generator connected to the device under test (DUT), with the output of the DUT connected to a spectrum analyser:

$$\boxed{\textit{RF Signal Generator} \rightarrow \textit{DUT} \rightarrow \textit{Spectrum Analyser}} \tag{31}$$

A reference measurement is taken without the DUT wherever required. The generator power, cables, and adapters are kept unchanged when comparing the reference and DUT measurements.

B. Filter Transmission Measurement

The signal generator is first connected directly to the spectrum analyser using a short reference cable. The generator power is kept constant and the reference power P_{in} is measured over the required frequency range.

The filter is then inserted between the generator and spectrum analyser, and the output power P_{out} is measured at the same frequencies. The insertion loss is calculated using

$$IL = P_{\text{in}} - P_{\text{out}}. \tag{32}$$

The insertion loss is plotted against frequency to determine the filter type and its characteristic frequencies.

C. Filter Reflection and Return Loss

For the reflection measurement, the signal generator is connected to one port of an RF Tee, the oscilloscope to the second port, and the filter input to the third port. The filter output is terminated with a 50 Ω load.

The oscilloscope input impedance is set to 1 MΩ. The maximum and minimum voltages, $V_{\max}$ and $V_{\min}$, are measured at the Tee junction.

The reflection coefficient is estimated from

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}, \tag{33}$$

and the return loss is calculated as

$$RL = -20 \log_{10} |\Gamma|. \tag{34}$$

D. Quarter-Wave Stub Measurement

The signal generator and spectrum analyser are connected to two ports of an RF Tee, while an open-ended coaxial stub is connected to the third port.

The expected notch frequency is calculated from

$$f_0 = \frac{V_F c}{4L}. \tag{35}$$

The output power is first measured without the stub and then with the stub over the same frequency range. The attenuation is calculated as

$$A = P_{\text{in}} - P_{\text{out}}. \tag{36}$$

The minimum in the output-power response gives the measured notch frequency.

E. Mixer Characterization

The RF and LO ports of the mixer are connected to two RF signal generators, while the IF port is connected to the spectrum analyser.

For input frequencies f_{RF} and f_{LO}, the expected mixing products are

$$f_{\text{sum}} = f_{\text{RF}} + f_{\text{LO}}, \tag{37}$$

and

$$f_{\text{diff}} = |f_{\text{RF}} - f_{\text{LO}}|. \tag{38}$$

The measured IF power P_{IF} is recorded along with the RF input power P_{RF}. The conversion loss is

$$L_{\text{conv}} = P_{\text{RF}} - P_{\text{IF}}. \tag{39}$$

F. Power Splitter Measurement

The splitter input is connected to the signal generator. One output is connected to the spectrum analyser and the other is terminated with a 50 Ω load.

The input power P_{in} and output powers P_{out1} and P_{out2} are measured by interchanging the output connections.

The losses are calculated as

$$L_1 = P_{\text{in}} - P_{\text{out1}}, \tag{40}$$

$$L_2 = P_{\text{in}} - P_{\text{out2}}. \tag{41}$$

The measured values are compared with the ideal 3 dB power division of an equal two-way splitter.

G. Power Combiner Measurement

Two signal generators are connected to the two input ports of the RF combiner, and the combined output is connected to the spectrum analyser.

The generators are set to the required frequencies and power levels. The output spectrum is observed and the resulting frequency components and power levels are recorded.

For equal-frequency inputs, the effect of the relative phase is observed, while for different-frequency inputs, the corresponding frequency components are identified.

VI. OBSERVATIONS AND GRAPHS

A. Filter Characterization

The measured input power without the filter, output power with the filter, and corresponding insertion loss are given in Table I.

TABLE I: Measured transmission characteristics of the low-pass filter.

f (MHz)	P_{in} (dBm)	P_{out} (dBm)	IL (dB)
50	-1.63	-1.22	2.85
100	-1.63	-1.54	0.09
150	-1.63	-1.77	0.14
200	-1.63	-1.60	0.03
250	-1.63	-1.67	0.04
300	-1.63	-2.01	0.38
350	-1.63	-2.16	0.53
400	-1.63	-2.66	1.03
420	-1.63	-2.48	0.85
440	-1.63	-3.03	1.40
460	-1.63	-3.35	1.72
500	-1.63	-3.64	2.01
520	-1.63	-4.25	2.62
540	-1.63	-4.37	2.74
560	-1.63	-5.20	3.57
580	-1.63	-6.60	4.97
600	-1.63	-7.60	5.97
620	-1.63	-10.60	8.97
640	-1.63	-16.50	14.87
660	-1.63	-26.20	24.57
680	-1.63	-29.20	27.57
700	-1.63	-31.56	29.93
720	-1.63	-39.50	37.87
740	-1.63	-44.32	42.69

1. Observation

The insertion loss remains relatively small over the lower-frequency region, indicating the passband of the filter. The attenuation increases significantly above approximately 540–550 *MHz. The* -3 *dB cutoff frequency is therefore approximately (from graph)*

$$f_c \approx 545.27 \text{ MHz}. \tag{42}$$

The filter enters its strongly attenuating stopband at approximately (from graph)

$$f_s \approx 650.58 \text{ MHz}. \tag{43}$$

The maximum measured insertion loss in the data occurs at 740 *MHz, where it reaches approximately* 42.69 *dB. The measured response is therefore consistent with a low-pass filter.*

B. Filter Reflection and Return Loss

The measured voltage values and calculated reflection parameters are given in Table II.

TABLE II: Measured filter reflection parameters.

f (MHz)	V_{RMS} (mV)	$V_{\min}$ (mV)	$V_{\max}$ (mV)	V_{inc} (mV)	V_{ref} (mV)	$\lvert\Gamma\rvert$	RL (dB)
50	154.46	126.75	155.77	141.26	14.51	0.1027	19.77
220	126.75	–	–	–	–	–	–
268	155.77	–	–	–	–	–	–

1. Observation

At 50 *MHz, the measured reflection coefficient is*

$$|\Gamma| = 0.1027, \tag{44}$$

corresponding to a return loss of approximately

$$RL = 19.77 \text{ dB}. \tag{45}$$

This indicates relatively good impedance matching at the measured frequency.

C. Quarter-Wave Stub

The measured transmission response of the open-ended coaxial stub is given in Table III.

TABLE III: Measured response of the quarter-wave open-circuit stub.

f (MHz)	P_{in} (dBm)	P_{out} (dBm)	Attenuation (dB)
5	0	-0.93	0.93
10	0	-1.20	1.20
15	0	-1.59	1.59
20	0	-2.05	2.05
25	0	-2.56	2.56
30	0	-3.10	3.10
35	0	-3.75	3.75
40	0	-4.60	4.60
45	0	-6.10	6.10
50	0	-9.40	9.40
53	0	-13.60	13.60
54	0	-15.90	15.90
55	0	-19.00	19.00
56	0	-23.90	23.90
57	0	-23.00	23.00
58	0	-19.00	19.00
59	0	-16.00	16.00
60	0	-9.67	9.67
65	0	-6.80	6.80
70	0	-5.10	5.10
80	0	-4.00	4.00
85	0	-3.00	3.00
90	0	-2.20	2.20
100	0	-1.30	1.30

Note of correction: *I have considered a uniform wire loss of -1.63 dBm throughout the table, which is incorrect as I found doing later experiments that wire loss also depends on the relative frequencies concerned.*

1. Observation

The attenuation increases rapidly as the frequency approaches 56 *MHz. The minimum measured output power is*

$$P_{\text{out,min}} = -23.9 \text{ dBm}, \tag{46}$$

at

$$f_0 = 56 \text{ MHz}. \tag{47}$$

Hence, the experimentally observed notch frequency of the stub is approximately 56 *MHz. The attenuation at this frequency is approximately* 23.9 *dB. Beyond the notch, the attenuation decreases again, demonstrating the frequency-selective behaviour of the quarter-wave stub.*

D. Mixer Characterization

The measured RF and LO frequencies, expected and measured IF frequencies, and conversion loss are shown in Table IV.

TABLE IV: Measured mixer characteristics.

f_{RF} (MHz)	f_{LO} (MHz)	$f_{\text{IF,exp}}$ (MHz)	$f_{\text{IF,meas}}$ (MHz)	P_{IF} (dBm)	L_{conv} (dB)
30	10	20	19.99	-13.35	10.35
30	10	40	39.99	-13.37	10.37
40	10	50	49.97	-13.30	10.30
40	10	30	29.97	-13.50	10.50
50	10	60	59.99	-13.30	10.30
50	10	40	39.99	-13.20	10.20

1. Observation

The measured IF frequencies agree very closely with the expected sum and difference frequencies. The maximum frequency deviation is approximately 0.03 *MHz.*

The conversion loss varies from 10.20 *to* 10.50 *dB, with an average value of approximately*

$$L_{\text{conv,avg}} \approx 10.34 \text{ dB}. \tag{48}$$

The results confirm the frequency-conversion operation of the mixer.

E. Power Splitter

The measured input and output powers of the two-way splitter are shown in Table V.

TABLE V: Measured power splitter characteristics.

f (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	L_1, L_2 (dB)
50	0	-4.50	-4.63	4.50, 4.63
100	0	-4.50	-4.51	4.50, 4.51
150	0	-4.80	-4.80	4.80, 4.80

1. Observation

The measured losses of the two output ports are approximately 4.5–4.8 *dB. The two outputs are closely balanced, with only a small difference between the two output powers.*

For an ideal two-way splitter, the theoretical division loss is approximately 3.01 *dB. The additional measured loss indicates practical insertion losses within the splitter and measurement system.*

F. Power Combiner

The measured output powers for two equal input signals are given in Table VI.

TABLE VI: Measured power combiner characteristics.

f (MHz)	P_1 (dBm)	P_2 (dBm)	P_{out} (dBm)
50	-3	-3	-7.26
100	-3	-3	-7.41
150	-3	-3	-7.54

1. Observation

For equal input powers of -3 *dBm, the measured combined output power varies from* -7.26 *to* -7.54 *dBm over the measured frequency range. The output power decreases slightly as frequency increases, indicating a small frequency dependence in the practical combiner response.*

VII. GRAPHS AND RESULTS

A. Low-Pass Filter Response

The measured insertion loss as a function of frequency is shown in Fig. 1. The response remains relatively flat at low frequencies and increases rapidly beyond the cutoff region.

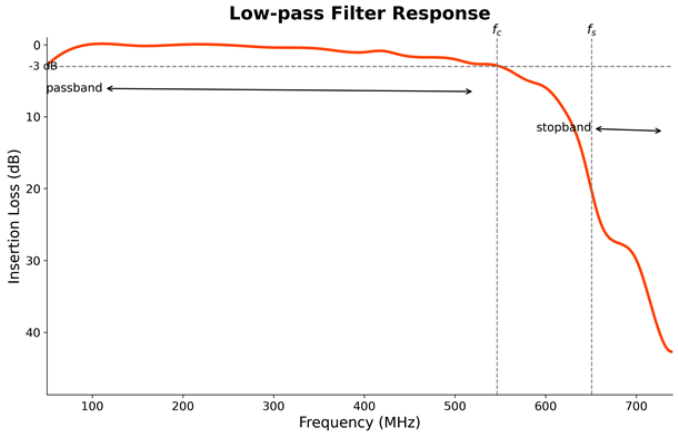


FIG. 1: Measured insertion-loss response of the low-pass filter.

From the measured response, the cutoff frequency is approximately

$$f_c \approx 545.27 \text{ MHz}, \tag{49}$$

while the stopband begins at approximately

$$f_s \approx 650.58 \text{ MHz}. \tag{50}$$

B. Quarter-Wave Stub Response

The measured output-power response of the open-ended stub is shown in Fig. 2. A pronounced minimum occurs at approximately 56 *MHz.*

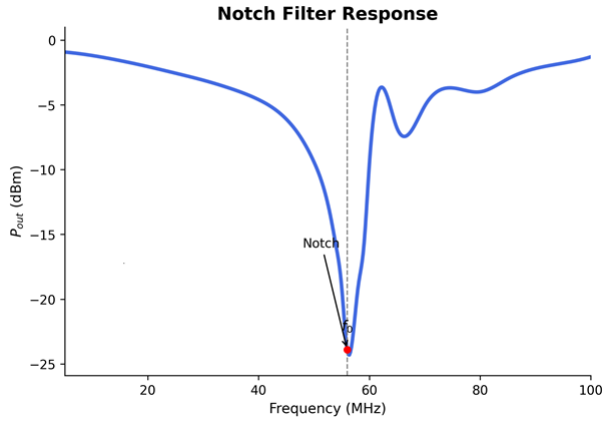


FIG. 2: Measured notch response of the quarter-wave open-circuit stub.

The minimum output power occurs at

$$f_0 \approx 56 \text{ MHz}, \tag{51}$$

where the measured attenuation is 23.9 *dB.*

VIII. OVERALL OBSERVATIONS

The experimental results demonstrate the frequency-selective and power-handling characteristics of the RF components investigated. The filter exhibits a low-pass response, with a cutoff frequency of approximately 545 *MHz and strong attenuation in the stopband. The filter reflection measurement gives a return loss of approximately* 19.77 *dB at* 50 *MHz.*

The quarter-wave stub produces a pronounced notch at approximately 56 *MHz, with a maximum measured attenuation of* 23.9 *dB. The mixer measurements show good agreement between the expected and measured sum and difference frequencies, with an average conversion loss of approximately* 10.34 *dB.*

The two-way splitter produces approximately 4.5–4.8 *dB loss at each output, while the combiner gives output powers between* −7.26 *and* −7.54 *dBm for equal* −3 *dBm input signals. The deviations from ideal values are attributable to the practical characteristics of the RF components and measurement setup.*

IX. CALCULATIONS AND ERROR ANALYSIS

A. Low-Pass Filter

The insertion loss of the filter is calculated using

$$IL = P_{\text{in}} - P_{\text{out}}, \tag{52}$$

where P_{in} *is the reference input power and* P_{out} *is the measured output power.*

The input power at the filter is obtained by accounting for the measured wire loss. The total wire loss was 1.63 *dB, while the power supplied by the signal generator was* 0 *dBm. Therefore,*

$$P_{\text{in}} = 0 - 1.63 = -1.63 \text{ dBm}. \tag{53}$$

For example, at 560 *MHz, the measured output power was*

$$P_{\text{out}} = -5.20 \text{ dBm}. \tag{54}$$

Hence,

$$IL = (-1.63) - (-5.20) = 3.57 \text{ dB}. \tag{55}$$

The cutoff frequency is determined from the −3 *dB point relative to the passband insertion loss. From the measured data in the passband, the insertion loss remains close to* 0 *dB. Therefore, the* −3 *dB cutoff level is approximately*

$$IL_{3\text{dB}} = IL_{\text{PB}} + 3 \text{ dB} \approx 0 + 3 = 3 \text{ dB}, \tag{56}$$

where IL_{PB} is the passband insertion loss.
From the measured data,

$$IL(540 \text{ MHz}) = 2.74 \text{ dB}, \tag{57}$$
$$IL(560 \text{ MHz}) = 3.57 \text{ dB}. \tag{58}$$

Since the 3 dB point lies between 540 MHz and 560 MHz, linear interpolation is used:

$$f_c = 540 + \frac{3.00 - 2.74}{3.57 - 2.74}(560 - 540), \tag{59}$$

giving

$$\boxed{f_c = 546.27 \text{ MHz}}. \tag{60}$$

The stopband frequency is determined at the 20 dB attenuation level. From the measured data,

$$IL(640 \text{ MHz}) = 14.87 \text{ dB}, \tag{61}$$
$$IL(660 \text{ MHz}) = 24.57 \text{ dB}. \tag{62}$$

Since the 20 dB point lies between these two frequencies, linear interpolation gives

$$f_s = 640 + \frac{20 - 14.87}{24.57 - 14.87}(660 - 640), \tag{63}$$

which gives

$$\boxed{f_s = 650.58 \text{ MHz}}. \tag{64}$$

Thus, the measured frequency response corresponds to a low-pass filter with a cutoff frequency of approximately 546.27 MHz and a stopband frequency of approximately 650.58 MHz.

B. Reflection Coefficient and Return Loss

The measured maximum and minimum voltages at 268 MHz and 220 MHz frequencies are respectively

$$V_{\max} = 155.77 \text{ mV}, \tag{65}$$
$$V_{\min} = 126.75 \text{ mV}. \tag{66}$$

The incident voltage is

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2} = \frac{155.77 + 126.75}{2} = 141.26 \text{ mV}. \tag{67}$$

The reflected voltage is

$$V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2} = \frac{155.77 - 126.75}{2} = 14.51 \text{ mV}. \tag{68}$$

Therefore, the magnitude of the reflection coefficient is

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{14.51}{141.26} = 0.1027. \tag{69}$$

The return loss is calculated from

$$RL = -20 \log_{10} |\Gamma|. \tag{70}$$

Thus,

$$RL = -20 \log_{10}(0.1027) \approx 19.77 \text{ dB}. \tag{71}$$

Hence,

$$\boxed{RL = 19.77 \text{ dB}}. \tag{72}$$

C. Quarter-Wave Open-Circuit Stub

For an open-circuit quarter-wave stub, the theoretical notch frequency is

$$f_0 = \frac{V_F c}{4L}, \tag{73}$$

where V_F is the velocity factor of the coaxial cable, c is the speed of light, and L is the physical length of the stub.
From the experimental setup,

$$V_F = 0.66, \tag{74}$$
$$c = 3 \times 10^8 \text{ m/s}, \tag{75}$$
$$L = 92.5 \text{ cm} = 0.925 \text{ m}. \tag{76}$$

Substituting these values,

$$f_0 = \frac{(0.66)(3 \times 10^8)}{4(0.925)}. \tag{77}$$

Therefore,

$$f_0 = 5.35 \times 10^7 \text{ Hz} \approx 53.5 \text{ MHz}. \tag{78}$$

Hence, the theoretical notch frequency is

$$\boxed{f_{0,\text{theory}} = 53.5 \text{ MHz}}. \tag{79}$$

From the measured frequency response, the minimum output power occurs at

$$\boxed{f_{0,\mathrm{exp}} = 56.0\ \mathrm{MHz}}, \tag{80}$$

with

$$\boxed{P_{\mathrm{out,min}} = -23.90\ \mathrm{dBm}}. \tag{81}$$

Since the input power was 0 *dBm, the maximum attenuation at the notch frequency is*

$$A_{\mathrm{max}} = P_{\mathrm{in}} - P_{\mathrm{out,min}}, \tag{82}$$

$$A_{\mathrm{max}} = 0 - (-23.90) = 23.90\ \mathrm{dB}. \tag{83}$$

Thus,

$$\boxed{A_{\mathrm{max}} = 23.90\ \mathrm{dB}}. \tag{84}$$

The difference between the theoretical and experimental notch frequencies is

$$\Delta f = f_{0,\mathrm{exp}} - f_{0,\mathrm{theory}} = 56.0 - 53.5 = 2.5\ \mathrm{MHz}. \tag{85}$$

The percentage error is

$$\%\mathrm{Error} = \frac{|f_{0,\mathrm{exp}} - f_{0,\mathrm{theory}}|}{f_{0,\mathrm{theory}}} \times 100. \tag{86}$$

Therefore,

$$\%\mathrm{Error} = \frac{|56.0 - 53.5|}{53.5} \times 100 \approx 4.67\%. \tag{87}$$

Hence,

$$\boxed{\%\mathrm{Error} = 4.67\%}. \tag{88}$$

D. Mixer

The expected intermediate-frequency components are obtained from the sum and difference of the RF and LO frequencies:

$$f_{\mathrm{sum}} = f_{\mathrm{RF}} + f_{\mathrm{LO}}, \tag{89}$$

$$f_{\mathrm{diff}} = |f_{\mathrm{RF}} - f_{\mathrm{LO}}|. \tag{90}$$

For example, for

$$f_{\mathrm{RF}} = 30\ \mathrm{MHz}, \tag{91}$$

$$f_{\mathrm{LO}} = 10\ \mathrm{MHz}, \tag{92}$$

the expected difference frequency is

$$f_{\mathrm{diff}} = |30 - 10| = 20\ \mathrm{MHz}. \tag{93}$$

The measured frequency was 19.99 *MHz, giving an error of*

$$\Delta f = 20 - 19.99 = 0.01\ \mathrm{MHz}. \tag{94}$$

The conversion loss is calculated using

$$L_{\mathrm{conv}} = P_{\mathrm{RF}} - P_{\mathrm{IF}}. \tag{95}$$

For the same measurement,

$$P_{\mathrm{RF}} = -3\ \mathrm{dBm}, \tag{96}$$

$$P_{\mathrm{IF}} = -13.35\ \mathrm{dBm}. \tag{97}$$

Thus,

$$L_{\mathrm{conv}} = (-3) - (-13.35) = 10.35\ \mathrm{dB}. \tag{98}$$

For the six measurements, the conversion losses were

$$10.35,\ 10.37,\ 10.30,\ 10.50,\ 10.30,\ 10.20\ \mathrm{dB}. \tag{99}$$

The average conversion loss is therefore

$$L_{\mathrm{conv,avg}} = \frac{10.35 + 10.37 + 10.30 + 10.50 + 10.30 + 10.20}{6}, \tag{100}$$

giving

$$\boxed{L_{\mathrm{conv,avg}} \approx 10.34\ \mathrm{dB}}. \tag{101}$$

E. Power Splitter

The loss at each output port of the two-way splitter is calculated using

$$L = P_{\mathrm{in}} - P_{\mathrm{out}}. \tag{102}$$

At 50 *MHz, the input power is*

$$P_{\mathrm{in}} = 0\ \mathrm{dBm}. \tag{103}$$

For output port 1,

$$P_{\text{out1}} = -4.50 \text{ dBm}, \tag{104}$$

and hence

$$L_1 = 0 - (-4.50) = 4.50 \text{ dB}. \tag{105}$$

For output port 2,

$$P_{\text{out2}} = -4.63 \text{ dBm}, \tag{106}$$

giving

$$L_2 = 0 - (-4.63) = 4.63 \text{ dB}. \tag{107}$$

For an ideal two-way equal splitter, the theoretical division loss is

$$L_{\text{ideal}} = 10 \log_{10}(2) = 3.01 \text{ dB}. \tag{108}$$

The measured losses are therefore slightly greater than the ideal value, indicating additional practical insertion losses in the splitter and measurement system.

F. Power Combiner

For the combiner experiment, both input signals have a power of

$$P_1 = P_2 = -3 \text{ dBm}. \tag{109}$$

The total input power is obtained by first converting the individual powers to linear units:

$$P_{\text{total}} = 2 \times 10^{-3/10} \text{ mW}. \tag{110}$$

In dBm,

$$P_{\text{total,dBm}} = 10 \log_{10} \left(2 \times 10^{-3/10} \right) \approx 0.01 \text{ dBm}. \tag{111}$$

At 50 MHz, the measured combined output power is

$$P_{\text{out}} = -7.26 \text{ dBm}. \tag{112}$$

Therefore, the measured loss is

$$L = P_{\text{total,dBm}} - P_{\text{out}}, \tag{113}$$

$$L = 0.01 - (-7.26) = 7.27 \text{ dB}. \tag{114}$$

Hence,

$$\boxed{L \approx 7.27 \text{ dB}}. \tag{115}$$

Note of correction :*We cannot term L as $L_{combiner}$ as we do not know the phase of the input signals, so we do not know if they are combining or subtracting.*

X. RESULTS

The RF components were characterized through measurements of transmission, reflection, frequency conversion, and power division. The filter exhibited a low-pass response with a cutoff frequency of 546.27 MHz and a 20 dB stopband frequency of 650.58 MHz. The measured return loss at 50 MHz was 19.77 dB, indicating relatively good input impedance matching.

For the quarter-wave open-circuit stub, the theoretical notch frequency was calculated as 53.5 MHz, while the experimental notch occurred at 56.0 MHz. The corresponding percentage difference was 4.67%. The minimum measured output power at the notch was −23.90 dBm, corresponding to an attenuation of 23.90 dB.

The mixer measurements showed good agreement between the expected and measured frequency-conversion products, with an average conversion loss of approximately 10.34 dB. The two-way splitter showed output losses of approximately 4.50–4.63 dB at 50 MHz, compared with the ideal division loss of 3.01 dB. For the power combiner, the measured loss at 50 MHz was approximately 7.27 dB.

XI. CONCLUSION

- ***Filter characterization:*** *The measured response showed that the filter behaves as a low-pass filter. The signal passes with relatively low loss at lower frequencies, while the attenuation increases strongly after the cutoff region. The measured cutoff frequency was $f_c = 546.27$ MHz, and the 20 dB stopband frequency was $f_s = 650.58$ MHz. This confirms the expected frequency-selective behaviour of a low-pass filter.*

  The filter successfully passes lower-frequency signals and strongly attenuates frequencies above approximately 546 MHz.

- ***Filter reflection and return loss:*** *The reflection measurement at 50 MHz gave a reflection coefficient of $|\Gamma| = 0.1027$. This corresponds to a return loss of approximately 19.77 dB. A higher return loss means that a smaller fraction of the incident signal is reflected back towards the source. The result therefore indicates reasonably good impedance matching at the measured frequency.*

- ***Quarter-wave stub:*** *The theoretical notch frequency of the open-circuit quarter-wave stub was calculated using the measured velocity factor and stub length. The theoretical value was* 53.5 *MHz, while the experimental minimum occurred at* 56.0 *MHz. The difference between the two values gives a percentage error of approximately* 4.67%. *At the measured notch frequency, the output power reached* −23.90 *dBm from an input of* 0 *dBm, giving a maximum attenuation of* 23.90 *dB.*

  The stub produced a clear notch near the predicted frequency, with only 4.67% *difference between theory and experiment.*

- ***Mixer characterization:*** *The mixer produced the expected sum and difference frequency components when RF and LO signals were applied. The measured IF frequencies were very close to the calculated values, showing that the mixer performed the expected frequency conversion. The measured conversion losses were close to each other, with an average value of approximately* 10.34 *dB.*

  The mixer successfully converted the input frequencies, and the average measured conversion loss was about 10.34 *dB.*

- ***Power splitter and voltage–power relation:*** *The splitter divided the input signal between its two output ports. The measured losses were approximately* 4.5–4.8 *dB, which is slightly higher than the ideal* 3.01 *dB loss for an equal two-way splitter. This extra loss is due to the practical losses of the splitter and measurement setup.*

  The experiment also helps demonstrate the relation between voltage and power. For a fixed resistance,

$$P = \frac{V_{\text{rms}}^2}{R}, \tag{116}$$

so that

$$P \propto V^2. \tag{117}$$

For equal power division, each output receives half of the input power. Therefore,

$$\frac{P_{\text{out}}}{P_{\text{in}}} = \frac{1}{2}, \tag{118}$$

and the corresponding voltage ratio is

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \sqrt{\frac{1}{2}} = 0.707. \tag{119}$$

  Thus, a measured voltage of 0.707 *V is consistent with an output voltage equal to* 70.7% *of the input voltage (1V), when the impedance remains the same.*

  Halving the power does not halve the voltage; instead, the voltage becomes 0.707 *times the input voltage.*

- ***Power combiner:*** *The power combiner performs the opposite function of a splitter by bringing two input signals together at one output. The output is not determined only by the input powers; it also depends on the frequencies and relative phases of the two signals. Signals with the same frequency can add constructively or cancel partly depending on their phase, while signals at different frequencies appear as separate components in the output spectrum.*

  The measured output therefore demonstrates that a practical combiner must be treated as both a power-combining and phase-dependent device.

  A combiner adds signals together, but the final output depends strongly on their frequency, amplitude, and relative phase.

- ***Measurement uncertainties:*** *Small differences between the theoretical and measured results are expected in practical RF measurements. Possible sources include cable and connector losses, impedance mismatch, instrument accuracy, frequency and power reading errors, and non-ideal behaviour of the RF components. These effects can influence the measured power, voltage, and frequency values.*

XII. REFERENCES

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2. *Wikipedia*
3. *Components and Measurements, YouTube video playlist: YouTube instructional video playlist.*
4. *Datasheet of RG174 -Datasheet*
5. *AI tool for more conceptual understanding.*

XIII. DECLARATION OF AI USAGE

I have used AI tools in the following ways for the completion of my lab report:

- *Formatting the theory content into various sub sections and concise writing of the same after I studied them from various sources. It helped me compile them well.*

- *Formatting of graph in python (lines along with labels) and the algorithm to find the cut off frequency from the graph.*

However ,the calculations, error analysis, data acquisition and processing, conclusions, key takeaways, results were all self written from scratch and there is absolutely no AI aid taken over those sections.

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